CASE STUDY — Severe Weather Risk Analytics

A U.S.-based insurance and risk-management company wants to understand how severe weather is affecting different states across the country.

The company regularly reviews storm-related events such as thunderstorms, tornadoes, floods, hail, strong winds and other severe weather conditions. At present, the analysis is mostly performed using downloaded files and spreadsheets. As the amount of historical data grows, it has become difficult to maintain the files, compare one year with another and prepare consistent reports.

The company has decided to move this analysis into Snowflake.

Your team has been asked to build a simple data pipeline that can take yearly storm-event files, store the original information, prepare it for analysis and finally make the required information available for reporting.

Data Source

Use the NOAA Storm Events raw Details files.

Initial files

2022

```
https://www.ncei.noaa.gov/pub/data/swdi/stormevents/csvfiles/StormEvents_details-ftp_v1.0_d2022_c20260625.csv.gz
```

2023

```
https://www.ncei.noaa.gov/pub/data/swdi/stormevents/csvfiles/StormEvents_details-ftp_v1.0_d2023_c20260323.csv.gz
```

2024

```
https://www.ncei.noaa.gov/pub/data/swdi/stormevents/csvfiles/StormEvents_details-ftp_v1.0_d2024_c20260728.csv.gz
```

Use these three files for the initial implementation.

After the complete pipeline is working, use the following file as a new incoming data file.

2025

```
https://www.ncei.noaa.gov/pub/data/swdi/stormevents/csvfiles/StormEvents_details-ftp_v1.0_d2025_c20260728.csv.gz
```

Treat every year as a separate source file.

Do not manually combine the files before loading them.

Business Requirement

Management wants to understand which parts of the United States are more frequently affected by severe weather and which weather events create greater business risk.

They are particularly interested in understanding whether some states are repeatedly affected, whether particular types of storms cause greater property or crop damage and whether weather activity is increasing from year to year.

The final solution should allow management to answer questions such as:

- Which states have the highest number of storm events?
- Which storm events occur most frequently?
- Which states experience the highest property damage?
- Which states experience the highest crop damage?
- Which types of events create the greatest financial impact?
- Which months experience more severe-weather activity?
- How does storm activity change between 2022, 2023 and 2024?
- Are particular storm types more common in certain states?
- Which states appear to have the highest overall weather-related risk?
- What changes when the 2025 data becomes available?

The team should inspect the source data and decide which information is required to answer these questions.

Work to Be Completed

Step 1 — Understand the Source Data

Start by downloading and reviewing the 2022 NOAA file.

Understand what one record represents and review the information available in the file.

Identify the fields that could help answer the management questions.

Do not assume that every field in the source needs to be used.

The team should understand the business meaning of the data before designing the solution.

Step 2 — Plan the Data Flow

Before building anything, prepare a simple architecture showing how the data will move from the NOAA source into Snowflake and finally into the reporting layer.

The design should clearly show:

NOAA Source Files

↓

Landing Location

↓

Snowflake Ingestion

↓

Raw Data

↓

Cleaned Data

↓

Reporting Data

↓

Power BI

The team should decide which Snowflake services or objects are appropriate for each stage.

Be prepared to explain why each component is required.

Step 3 — Prepare the Landing Area

Download the 2022, 2023 and 2024 files and place them in the location selected for incoming data.

Treat each file as an independent arrival.

The company wants the same process to work when future yearly files are received.

The landing area should therefore not be designed only for these three files.

Step 4 — Ingest the Raw Files into Snowflake

Create the required Snowflake setup to access and ingest the files.

Load the raw information with minimum business transformation.

The company wants to retain the source information so that records can later be traced back to the original file.

The raw layer should therefore make it possible to identify:

- where the record came from;
- when it entered Snowflake;
- which source file contained it.

Do not perform all business cleaning directly during the raw ingestion stage.

Step 5 — Review and Clean the Data

Once the raw data is available, investigate the quality of the source.

Look for conditions such as:

- missing values;
- duplicate records;
- inconsistent dates;
- unusual numeric values;
- missing state or event information;
- damage values that need interpretation;
- records that may require standardisation.

Do not remove data simply because something looks unusual.

Understand the meaning first and decide how it should be handled.

Create a cleaned layer that is suitable for further business analysis.

Step 6 — Create Business-Friendly Information

The raw source may not contain every value in the exact form management wants to see.

Prepare useful business information from the cleaned data.

For example, management may need to understand:

- year;
- month;
- state;
- storm category;
- overall damage;
- level of financial impact;
- event frequency;
- severity of an event.

The team should decide which derived values are required.

At least one repeated piece of business logic should be implemented in a reusable way using Snowflake.

Step 7 — Implement Incremental Processing

The company does not want all historical data to be processed again whenever a new yearly file arrives.

After the 2022, 2023 and 2024 data has been completely processed, introduce the 2025 file.

The solution should identify the newly arrived information and process only what is required.

Previously processed records should not unnecessarily move through the complete pipeline again.

The team should demonstrate evidence that the 2025 file was handled as a new arrival.

Step 8 — Prepare the Reporting Layer

Create a reporting-ready layer based on the business questions.

The reporting layer should make it easy to analyse storm events by areas such as:

- state;
- event type;
- year;
- month;
- damage;
- severity.

Do not connect Power BI directly to the raw source table.

The team should decide how the reporting tables should be structured.

Step 9 — Build the Business Report

Connect Power BI to the final Snowflake reporting layer.

Management should be able to understand:

Overall Weather Activity

- total number of storm events;
- overall property damage;
- overall crop damage;
- yearly change in storm activity.

State-Level Risk

- states with the highest number of events;
- states with the greatest financial impact;
- states repeatedly affected by severe weather.

Event Analysis

- most frequent storm types;
- storm types causing the highest damage;
- event behaviour across different states.

Time Analysis

- storm activity by year;
- storm activity by month;
- changes in damage over time.

The team should decide which visualisation is appropriate for each business question.

Step 10 — Present the End-to-End Flow

At the end of the assignment, present the completed solution.

The presentation should show:

- the source data;
- the architecture;
- how the files are ingested;
- how raw data is retained;
- important cleaning decisions;
- business transformations;
- how incremental processing works;
- the final reporting structure;
- the Power BI report;
- and the main findings from the data.

The team should also demonstrate the arrival of the **2025 file** and explain what happens from the moment the new file reaches the landing location until the new information becomes available for reporting.
